



Vivo and Jio join hands

Vivo has teamed-up with Jio for its X80 smartphone series.
<https://digit.in/jun22-75>



New Star Wars game announced

Srat Wars Jedi: Survivor is the direct sequel to Star Wars Jedi: The Fallen Order.
<https://digit.in/jun22-76>

When going soft is harder

Encryption on SSDs is not such a great idea but there are solutions

Agent001 | agent001@digit.in



S

ince the early times that hardware has existed, it's always been the case that hardware-based

solutions offered a lot more resiliency than software-based ones. Encryption is one such thing. A hardware-based encryption mechanism, which might either be a separate chip handling the encryption processing or is simply a means for storing the private keys, has always been better than a software-based solution. This is why hardware password keys are so prevalent in movies and real life. Unfortunately, the way these things are implemented isn't always up to the mark.

Take, for example, Bitlocker. It's one of the most popular encryption services out there, not because it's perfect but because it's ubiquitous. Every Windows 7 or 10 or 11 machine with a 'Pro' license comes with support for Bitlocker. And using Bitlocker, one can encrypt every piece of data on their storage devices. Unless, of course, you have an SSD.

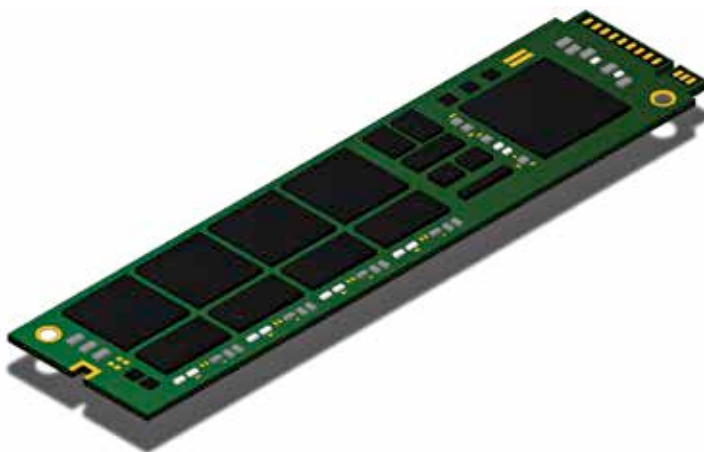
Researchers who have reverse-engineered the firmware on many SSDs found that hardware-based encryption is implemented in the most half-hearted manner possible on many popular brands for SSDs. The way Bitlocker functions in Windows 10 is that if it detects that

the SSD has hardware-based encryption capabilities, as is the case with most SSDs these days, then the job of encrypting the data is to be done by the SSD itself.

This is totally fine if the SSDs were to implement proper encryption but unfortunately, that's where the problem lies. Researchers found that there were many critical security weaknesses when it came to how the

firmware of the SSD and modify it before flashing it back on the SSD with clear passwords.

The researchers estimate that more than 60 per cent of the SSDs in the market, including the top brands, are guilty of having poor security implementation practices. In recent times, several SSD manufacturers are opting to remove hardware-based encryption capabilities from the SSDs



SSDs implemented encryption. In some cases, the researchers simply had to open the SSDs and tap into some of the debugging contact pads on the PCB and they could modify the password validation routine. This way, no matter what you entered, the password would always validate and the drive could be decrypted. In some cases, the researchers could read the

altogether. This also has an impact on performance since running Bitlocker easily drops read and write speeds by anywhere between 500-1000 MBps. So manufacturers have an incentive to cut costs and ensure that their drives perform better than competing drives in the SSD space.

So what's the solution? The good old software-based encryption! The